SIMATS ENGINEERING

LAB EXPERIMENTS

COURSE NAME: DATABASE MANAGEMENT SYSTEMS

COURSE CODE: CSA05

EXP 5 : TCL Commands – COMMIT, SAVEPOINT, ROLLBACK

```
mysql> create database BankingDB;
Query OK, 1 row affected (0.06 sec)

mysql> use bankingDB;
Database changed
mysql> CREATE TABLE Accounts (Acc_ID INT PRIMARY KEY, Acc_Name VARCHAR(50), Balance INT);
Query OK, 0 rows affected (0.08 sec)

mysql> DESC Accounts;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| Acc_ID   | int         | NO   | PRI | NULL    |       |
| Acc_Name | varchar(50) | YES  |     | NULL    |       |
| Balance  | int         | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
3 rows in set (0.06 sec)

mysql> INSERT INTO Accounts VALUES (401, 'Karthik', 15000), (402, 'Sneha', 22000), (403, 'Varun', 18000);
Query OK, 3 rows affected (0.06 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Accounts;
+--------+----------+---------+
| Acc_ID | Acc_Name | Balance |
+--------+----------+---------+
|    401 | Karthik  |   15000 |
|    402 | Sneha    |   22000 |
|    403 | Varun    |   18000 |
+--------+----------+---------+
3 rows in set (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE Accounts SET Balance = 20000 WHERE Acc_ID = 401;
Query OK, 1 row affected (0.04 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> SELECT * FROM Accounts;
+--------+----------+---------+
| Acc_ID | Acc_Name | Balance |
+--------+----------+---------+
|    401 | Karthik  |   20000 |
|    402 | Sneha    |   22000 |
|    403 | Varun    |   18000 |
+--------+----------+---------+
3 rows in set (0.00 sec)

mysql> SAVEPOINT S1;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE Accounts SET Balance = 25000 WHERE Acc_ID = 402;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM Accounts;
+--------+----------+---------+
| Acc_ID | Acc_Name | Balance |
+--------+----------+---------+
|    401 | Karthik  |   20000 |
|    402 | Sneha    |   25000 |
|    403 | Varun    |   18000 |
+--------+----------+---------+
3 rows in set (0.00 sec)

mysql> ROLLBACK TO S1;
Query OK, 0 rows affected (0.05 sec)

mysql> SELECT * FROM Accounts;
+--------+----------+---------+
| Acc_ID | Acc_Name | Balance |
+--------+----------+---------+
|    401 | Karthik  |   20000 |
|    402 | Sneha    |   22000 |
|    403 | Varun    |   18000 |
+--------+----------+---------+
3 rows in set (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> SELECT * FROM Accounts;
+--------+----------+---------+
| Acc_ID | Acc_Name | Balance |
+--------+----------+---------+
|    401 | Karthik  |   20000 |
|    402 | Sneha    |   22000 |
|    403 | Varun    |   18000 |
+--------+----------+---------+
3 rows in set (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE Accounts SET Balance = 10000 WHERE Acc_ID = 403;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> ROLLBACK;
Query OK, 0 rows affected (0.07 sec)

mysql> SELECT * FROM Accounts;
+--------+----------+---------+
| Acc_ID | Acc_Name | Balance |
+--------+----------+---------+
|    401 | Karthik  |   20000 |
|    402 | Sneha    |   22000 |
|    403 | Varun    |   18000 |
+--------+----------+---------+
3 rows in set (0.00 sec)
```